opportunity cost of time is the average after tax earnings. We estimate the income tax rate for a median household to be 7% and apply to 2021 ACS county-specific mean wage estimates. The income tax rate of 7% is the percentage difference in median post-tax income and median income from (U.S. Census Bureau, 2022). The probability that a parent is working is measured with the employment population ratio among people with their own children under 18 and is 80.7% (US Bureau of Labor Statistics, 2025). We apply the employment population ratio to the proportion of the total workforce that are employed parents. The average of this county-specific measure across all counties is approximately $239 (2015$).

To measure the loss of learning, we update findings of Gottfried & Ansari (2022), Aucejo & Romano (2016), and Liu et al. (2021). Gottfried & Ansari (2022) estimated the impact of school absences on learning as measured by end-of-grade math test scores for grades K-1. They found a one standard deviation increase in absences for a single student reduced that student's math test score by 0.11 standard deviations. We convert this result to estimate the effect on math test scores from a single absence by dividing by the standard deviation of absences, finding that a single absence results in a decrease in math test score of 0.023 standard deviations. Aucejo & Romano (2016) estimated the same for grades 3-5 and found a decrease of 0.006 standard deviations. Similarly, Liu et al. (2021) found a decrease of 0.004 standard deviations for grades 6-11.

We multiply the estimates of the effect of school absences on learning by an estimate of the effect of learning on lifetime earnings to calculate the effect of a single absence on lifetime earnings.

For BenMAP application, we create valuation functions that use the sum of caregiver costs and loss of learning for elementary school, middle school, and high school students. We estimate county-specific averages of the combined effect by summing the county-specific estimates of caregiver costs and age-specific costs per one lost school day with 2%, 3%, and 7% discounting. The average of this county-specific measure across all counties is approximately $3,817 (discounted at 2%), $2,831 (discounted at 3%) and $1,026 (discounted at 7%) per school day for ages 5 to 6 (2015$, Table H-14). For ages 7-12, the average is approximately $996 (discounted at 2%), $738 (discounted at 3%) and $268 (discounted at 7%) per school day. Finally, for ages 13-18, the average is approximately $664 (discounted at 2%), $492 (discounted at 3%) and $178 (discounted at 7%) per school day.

A unit value based on the approach described above is likely to understate the value of a school loss day in at least three ways:

1) It omits WTP to avoid the symptoms/illness which resulted in the school absence

2) The approach omits other aspects of school attendance such as social and emotional development or meals

3) It does not account for deleterious effects on student learning in other subjects.

H.5 Developing Income Growth Adjustment Factors

Chapter 4 of the BenMAP-CE User Manual provides instructions for formatting and adding income growth data. These values are used to adjust WTP estimates for growth in real income. As discussed in that chapter, evidence and theory suggest that WTP should increase as real income increases. When reviewing the economic literature to develop income growth adjustment factors, it is important to have an economist assist. For an overview of valuation, see Chapter 7: Aggregating, Pooling, and Valuing.

Adjusting WTP to reflect growth in real income requires three steps:

1. *Identify relevant income elasticity estimates from the peer-reviewed literature.*
2. *Calculate changes in future income.*
3. *Calculate adjustments to WTP based on changes in future income and income elasticity estimates.*

<u>1. Identifying income elasticity estimates</u>

Income elasticity estimates relate changes in demand for goods to changes in income. Positive income elasticity suggests that as income rises, demand for the good also rises. Negative income elasticity suggests that as income rises, demand for the good falls. BenMAP-CE does not adjust Cost-of-Illness (COI) estimates according to changes in income elasticity due to the fact that COI estimates the direct cost of a health outcome; instead we adjust this metric using inflation factors described above. BenMAP-CE includes income elasticity estimates specific to the type of health endpoint associated with the WTP estimate. BenMAP-CE contains elasticity estimates for three types of health effects: minor, severe and mortality. Minor health effects are those of short duration. Severe, or chronic, health effects are of longer duration. Consistent with economic theory, the peer reviewed literature indicates that income elasticity varies according to the severity of the health effect. A review of the literature revealed a range of income elasticity estimates that varied across the studies and according to the severity of health endpoint. Table H-15 summarizes the income elasticity estimates for minor health effects, severe health effects and mortality. Here we have provided a lower-, upper- and central-elasticity estimate for each type of health endpoint.

Table H-15. Income Elasticity Estimates

Health Endpoint	Lower Bound	Central Estimate	Upper Bound
Minor Health Effect	0.04	0.15	0.30
Severe and Chronic Health Effects	0.25	0.45	0.60
Mortality	0.08	0.40	1.00

2. Calculating changes in future income

The next input to the WTP adjustment is annual changes in income. Historical US Gross Domestic Product (GDP) data (1990-2016) comes from the U.S. Bureau of Commerce's Bureau of Economic Analysis (BEA). GDP values were adjusted for inflation using the BEA's price index for GDP. We divided historical GDP values by populations provided by the BEA to estimate GDP per capita to maintain internal consistency in the calculation. Future changes in annual income are based on data presented in the Annual Energy Outlook (AEO) 2020, a report prepared by the U.S. Energy Information Administration (EIA) (AEO, 2020). AEO published annual GDP projections through the year 2050, which were adjusted for inflation using the GDP Chain-type Price Index reported by AEO. We divided projected GDP values by AEO's population projections to estimate per capita GDP, again maintaining internal consistency in the calculation.

3. Calculating changes in WTP

The income elasticity estimates from Table H-15 and the estimated changes in future income may then be used to estimate changes in future WTP for each health endpoint. The adjustment formula follows four steps:

1)
$$\varepsilon = \frac{\frac{\Delta WTP}{WTP}}{\frac{\Delta I}{I}} = \frac{(WTP_2 - WTP_1) \times (I_2 + I_1)}{(I_2 - I_1) \times (WTP_2 + WTP_1)}$$

2)
$$\varepsilon I_2 WTP_2 + \varepsilon I_2 WTP_1 - \varepsilon I_1 WTP_2 - \varepsilon I_1 WTP_1 = I_2 WTP_2 + I_1 WTP_2 - I_2 WTP_1 - I_1 WTP_1$$

3)
$$WTP_2 \times (\varepsilon I_2 - \varepsilon I_1 - I_2 - I_1) = WTP_1 \times (\varepsilon I_1 - \varepsilon I_2 - I_1 - I_2)$$

4)
$$WTP_2 = WTP_1 \times \frac{\varepsilon I_1 - \varepsilon I_2 - I_1 - I_2}{\varepsilon I_2 - \varepsilon I_1 - I_2 - I_1}$$

Table H-16 summarizes the income-based WTP adjustments used within BenMAP-CE for minor health endpoints, severe health endpoints, and premature mortality. BenMAP-CE applies the "mid" income growth adjustment to the WTP for each corresponding health endpoint. The "low" and "upper" are provided for bounding the "mid" estimate.

Table H-16. Income-Based WTP Adjustments by Health Effect and Year

	Minor Health Endpoint			Severe Health Endpoint			Mortality		
Year	Low	Mid	Upper	Low	Mid	Upper	Low	Mid	Upper
1990	1	1	1	1	1	1	1	1	1
1991	0.999425	0.997845	0.995695	0.996411	0.99355	0.991409	0.99885	0.994264	0.985722
1992	1.000278	1.001043	1.002086	1.001738	1.003131	1.004177	1.000556	1.002783	1.006971
1993	1.000845	1.003171	1.006353	1.005291	1.009545	1.012746	1.00169	1.00848	1.021335
1994	1.001941	1.007299	1.014651	1.012194	1.022057	1.029519	1.003886	1.019583	1.049686
1995	1.002529	1.009516	1.019122	1.01591	1.028821	1.038614	1.005064	1.025578	1.065196
1996	1.003545	1.01336	1.026899	1.022366	1.040622	1.054532	1.007103	1.036027	1.092569
1997	1.004809	1.018155	1.036642	1.030442	1.055472	1.074652	1.009642	1.049157	1.127596
1998	1.006098	1.023062	1.046661	1.038734	1.070818	1.095552	1.012234	1.062703	1.164493
1999	1.007498	1.02841	1.057639	1.047803	1.087723	1.1187	1.015052	1.077598	1.205983
2000	1.008676	1.032928	1.066959	1.055489	1.102148	1.138556	1.017428	1.090286	1.242108
2001	1.008675	1.032924	1.06695	1.055482	1.102134	1.138537	1.017426	1.090274	1.242075
2002	1.008984	1.03411	1.069403	1.057504	1.105943	1.143797	1.018048	1.093621	1.251727
2003	1.009739	1.037019	1.075433	1.062468	1.115325	1.15678	1.019574	1.101858	1.275708
2004	1.010863	1.041352	1.084451	1.069884	1.129408	1.176348	1.021844	1.114208	1.312269
2005	1.011864	1.045228	1.092549	1.076534	1.142112	1.194079	1.02387	1.125332	1.345838
2006	1.012602	1.04809	1.09855	1.081456	1.151559	1.207315	1.025363	1.133594	1.371175
2007	1.012957	1.049471	1.101452	1.083835	1.156139	1.213747	1.026084	1.137596	1.383575
2008	1.012532	1.047821	1.097984	1.080992	1.150667	1.206063	1.025223	1.132814	1.368769
2009	1.011166	1.042525	1.086897	1.071894	1.13324	1.181689	1.022457	1.117566	1.322336
2010	1.011843	1.045146	1.092377	1.076393	1.141841	1.1937	1.023827	1.125095	1.345117
2011	1.012165	1.046395	1.094994	1.07854	1.145958	1.199463	1.024479	1.128697	1.356115
2012	1.012764	1.04872	1.099873	1.082541	1.153646	1.210245	1.025692	1.135418	1.376817
2013	1.013213	1.050467	1.103547	1.085551	1.159447	1.2184	1.026602	1.140487	1.392581
2014	1.013911	1.053184	1.109275	1.090241	1.168516	1.231181	1.028017	1.148402	1.417472
2015	1.014754	1.056473	1.116227	1.095927	1.179559	1.246797	1.029727	1.158031	1.4482
2016	1.015112	1.057873	1.119194	1.098352	1.184283	1.253496	1.030454	1.162147	1.461489
2017	1.015291	1.058571	1.120676	1.099562	1.186645	1.25685	1.030817	1.164204	1.468167
2018	1.016092	1.061712	1.127353	1.105013	1.197312	1.272029	1.032446	1.173487	1.498591
2019	1.016945	1.06506	1.134495	1.110837	1.208762	1.288383	1.03418	1.183439	1.531756
2020	1.017494	1.067221	1.139119	1.114604	1.216197	1.299038	1.035298	1.189894	1.553577
2021	1.017858	1.068652	1.142185	1.1171	1.221138	1.306133	1.036037	1.194181	1.568204
2022	1.018224	1.070097	1.145287	1.119624	1.226144	1.313335	1.036784	1.198523	1.583131
2023	1.018585	1.071523	1.148352	1.122117	1.231098	1.320474	1.03752	1.202817	1.598006
2024	1.018991	1.073127	1.151804	1.124923	1.236689	1.328546	1.038347	1.20766	1.61492
2025	1.019446	1.074925	1.155682	1.128074	1.242981	1.337648	1.039273	1.213106	1.634116
2026	1.019915	1.076785	1.159702	1.131337	1.249515	1.347122	1.040231	1.218759	1.654237
2027	1.02041	1.078749	1.163955	1.134787	1.256443	1.357191	1.041242	1.224747	1.675778
2028	1.020937	1.080842	1.168495	1.138469	1.263858	1.367993	1.042318	1.231152	1.699074
2029	1.021436	1.082827	1.17281	1.141964	1.270919	1.378307	1.043337	1.237246	1.721494
2030	1.021959	1.08491	1.17735	1.145639	1.278367	1.389214	1.044406	1.243669	1.745395
2031	1.022479	1.086985	1.181879	1.149303	1.285815	1.400148	1.045469	1.250087	1.76956
2032	1.022969	1.088943	1.186166	1.152768	1.292879	1.410546	1.046472	1.256169	1.792727
2033	1.023487	1.091016	1.190711	1.15644	1.300389	1.421627	1.047532	1.262629	1.817622
2034	1.024004	1.093088	1.195265	1.160115	1.307929	1.432783	1.048592	1.269111	1.842903
2035	1.024493	1.095052	1.199589	1.163603	1.315106	1.443429	1.049594	1.275275	1.867234
2036	1.02498	1.097006	1.203902	1.16708	1.322282	1.454101	1.050592	1.281434	1.891828
2037	1.025478	1.099009	1.208332	1.170647	1.329668	1.465113	1.051613	1.287768	1.917424
2038	1.025984	1.101051	1.212857	1.174289	1.33723	1.476418	1.052652	1.294248	1.943931
2039	1.026492	1.103099	1.217405	1.177947	1.34485	1.487839	1.053694	1.300771	1.970954
2040	1.027004	1.105171	1.222018	1.181654	1.352597	1.499482	1.054748	1.307398	1.998757
2041	1.027508	1.107212	1.226569	1.185309	1.36026	1.511031	1.055784	1.313948	2.026591
2042	1.02803	1.109326	1.231295	1.189101	1.368236	1.523084	1.056857	1.320759	2.055917
2043	1.028555	1.111457	1.236071	1.19293	1.376316	1.535329	1.057937	1.327653	2.086006
2044	1.029075	1.113573	1.240821	1.196736	1.384373	1.547576	1.059008	1.334521	2.116397
2045	1.029605	1.115728	1.245672	1.20062	1.392623	1.560152	1.060099	1.341548	2.147923
2046	1.030116	1.117812	1.250373	1.20438	1.400637	1.572403	1.061152	1.348368	2.178951
2047	1.030638	1.119946	1.255197	1.208236	1.408882	1.585045	1.06223	1.355378	2.211295
2048	1.031135	1.121978	1.259801	1.211913	1.416771	1.597176	1.063255	1.36208	2.242655
2049	1.03162	1.123965	1.264312	1.215513	1.424519	1.609122	1.064256	1.368656	2.273847
2050	1.032088	1.125883	1.268677	1.218994	1.432035	1.620744	1.065222	1.37503	2.304491